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Design and computational analysis of a metal hydride hydrogen storage system with hexagonal honeycomb based heat transfer enhancements-part A

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HIGHLIGHTS

- Designed a 50 kg La-based reactor with hexagonal heat transfer enhancement (HTE).
- Simulation studies show remarkably improved heat transfer and hydrogenation.
- Significant reduction in system weight with promising volumetric efficiency.
- Ninety per cent hydrogenation (approx. 600 g of H_2) achieved in 7200s.

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ABSTRACT

In this study, design and performance analysis is carried out for a 10 kWh metal hydride based hydrogen storage system. The system is equipped with distinctive aluminium hexagonal honeycomb based heat transfer enhancements (HTE) having higher surface area to volume ratio for effective heat transfer combined with low system weight addition. The system performance was studied under different operating conditions. The optimum absorption condition was achieved at 35 bar with water at room temperature as heat transfer fluid where up to 90% absorption was completed in 7200 s. The performance of the reactor was observed to significantly improve upon the addition of the HTE network at a minimal system weight penalty.

Introduction

In the wake of the on-going discourse on global warming, the efforts to explore and implement alternative sources of energy have been intensified. Globally, policies are being effected to increase the dependence on renewable energy. In this context, hydrogen has emerged as a promising option because of its high calorific value and benign environmental effects. However, hydrogen being essentially a carrier of energy requires

Abbreviations: MH, Metal hydride; ENG, Expanded natural graphite; HTF, Heat transfer fluid; PCM, Phase changing materials; HPMH, High pressure metal hydride; HTE, Heat Transfer Enhancement.

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Nomenclature

c	Fraction of hydrogen absorbed
M	Rate of absorption/desorption, kg/m^3s
P	Pressure, bar
T	Temperature, K
u	Velocity, m/s
R	Universal gas constant, J/mol/K
ΔS	Entropy of reaction, J/mol/K
ΔH	Enthalpy of reaction, J/mol
ε	Porosity
h	Heat transfer coefficient, W/m^2K
C_p	Specific heat capacity, J/kg/K
k	Thermal conductivity, W/m/K
Q	Heat source/sink, W/m^3
Subscripts	
eq	Equilibrium
f	Final
e	Effective
m	Metal
g	Gas
s	source/sink
a	Absorption
d	Desorption

an efficient storage mechanism. There are multiple methods of storing hydrogen, the chief being: high pressure gaseous storage, cryogenic liquid state storage and solid state storage in the form of a metal hydride. Of these solid state storage has attracted extensive attention, primarily because of its safe storage potential and high volumetric storage capacity. The storage mechanism for metal hydride hydrogen storage involves a reversible reaction between hydrogen gas and a metal or intermetallic compound.

$$M + 0.5xH_2 \Leftrightarrow MH_x + \Delta H$$

As can be observed, the hydrogenation reaction is exothermic whereas the reverse reaction i.e. the dehydrogenation reaction is endothermic. From the above equation, it is evident that the withdrawal of this heat from the reaction bed during hydrogenation and its supply during dehydrogenation plays an important role in the storage and subsequent release of hydrogen from a metal hydride based hydrogen storage system. Heat transfer from and to a metal hydride bed is a key factor influencing hydrogen storage in MH based systems.

Multiple studies have been conducted to highlight the importance of heat transfer in MH based hydrogen storage systems and to provide effective solutions for the same. Investigations were conducted to improve the heat transfer by introducing thermal conductivity enhancements such as metal shavings [1] and graphite compacts [2–5]. With the advent of sophisticated modelling software tremendous exploration in reactor design became possible. This resulted in design and optimization of various types of reactor configurations, which would have been almost impossible through experimentation. A great variety of designs have been studied, optimized and compared. This has added vastly to the understanding of hydrogen storage in metal hydrides-reaction kinetics, heat and mass transfer in the reacting beds, system development etc. Reactor designs with optimized fin dimensions [6–12] or optimum number of cooling tubes have been designed [12–21] which improve the heat transfer by providing additional area for heat transfer. Studies have explored the use of helical coils [22–25], heat pipes [26–28], phase change materials [29–32] etc.

More recently, Raju et al. developed a correlation for optimum number of embedded cooling tubes (ECT) in metal hydride hydrogen storage reactor [33]. On the basis of these correlations different reactor configurations for a 50 kg La-based MH storage reactor were studied with the help of 3-D mathematical models. The reactor with 6 inches nominal diameter with 99 ECT was found to be exhibit the optimum performance. The effect of various operating parameters such as supply pressure, heat transfer fluid flow rate, absorption temperature etc. on the storage reaction was also investigated. A comparative study on four different types of reactor configurations was undertaken by Sakreddine et al. [34]. The four different types of configurations studied were: a) Reference – a plain TiMn alloy bed b) Compacts – Ti–Mn alloy compacts stacked together c) Alternation – Ti–Mn alloy compacts with stainless steel fins in between and d) Compact-mixture – Compacts of a mixture of Ti–Mn alloy and stainless steel. It was observed that the arrangement comprising of compacts with fins in between i.e. Alternation was the most promising in terms of hydrogen storage capacity, absorption rate and ease of implementation. Chibani et al. studied the effect of various material properties on the performance of a MH reactor with hydrogen entering into the reactor through multiple pipes [35]. It was noted that the properties that most affected the hydrogen absorption process were: absorption rate constant, activation energy and thermal conductivity. A combination of plate fins fitted with cooling tubes as a thermal management system was proposed by Gkanas et al. [36]. A parametric study was conducted on the system where fin thickness, number of fins and the heat transfer coefficient were varied. The optimum number of fins and the optimum fin thickness was determined.

An interesting modification to plate fins for enhanced heat transfer was proposed by Chandra et al. [37]. Conical funnel shaped fins were used in conjunction with cooling tubes. A 5 kg $LaNi_5$ based system was equipped with 10, 13, and 19 fins with 2, 4 and 6 cooling tubes. It was observed that the 19 fins plus exhibited better performance that 10 fins combined with 4 cooling tubes. Lewis et al. proposed a MH reactor equipped with Embossed Plate Heat Exchanger and studied the performance of the reactor with the help of a mathematical model [38]. Different flow field designs such as parallel, pin-type and serpentine were investigated. Significant improvement in hydrogen absorption rates was not noted. Thermal coupling between a MH reactor and a 2.5 kW PEM fuel cell was investigated by Omrani et al. [39]. It was observed that total heat removal from the fuel cell stack was sufficient to supply heat to the MH reactors for catering to the PEMFC.

While the studies exploring heat transfer and performance of metal hydride reactors have been many and varied, these are mostly limited to lab-scale reactors containing at the maximum a few kg of alloy powder. The studies on large scale systems are few in number and mostly limited to simulation studies. In the present study, the design and performance analysis of a 10 kWh (approx. 50 kg alloy) tank is presented for back-up stationary power supply. A unique hexagonal honeycomb based heat transfer enhancement network has been used for improvement of the heat transfer characteristics of the system. The present simulation study is followed by an experimental study presented in part B of the article.

Reactor design

The reactor used in the present study is a 10 kWh cylindrical reactor containing approx. 50 kg of $La_{0.9}Ce_{0.1}Ni_5$. The reactor dimensions are fixed on the basis of system capacity, gravimetric capacity of the alloy selected and the maximum pressure within the reactor. On these bases, a standard pipe dimension of 6" nominal diameter is selected for the reactor body, yielding an inner diameter of 154.06 mm and outer diameter of 168.28 mm. The volume required to store the material including the expansion volume results in a total system height of 1100 mm.

Design of hexagonal honeycomb HTE

Cellular structures such as honeycombs, metal foams etc. are known for their use as heat transfer enhancements especially under conditions where system weight is a major constraint [40]. The surface area per unit volume (i.e., the surface area density or specific surface area) of a typical micro-cell honeycomb is about 3000 $m^2\ m^{-3}$, which makes it an ideal material for strengthening heat transfer [41,42]. A hexagonal honeycomb based heat transfer enhancement (HTE) network is used in the present study which has high structural efficiency and large surface area to volume ratio. Heat transfer surfaces used for MH reactors should have large surface area to volume ratio to ensure good heat transfer without adding to the parasitic system weight. Use of honeycomb based HTE network fulfils this objective. The dimensions have been optimized keeping in view the HTE strength and completion of conduction pathway from the reactor core to circumference. The former was done with the help of the relations presented by Gibson and Ashby [43] represented by Eqn. (1) for honeycombs under bi-axial loading where σ_1 and σ_2 are the stresses acting on the cell in direction x_1 and x_2 respectively, and σ_{fs} is the modulus of rupture of the material:

$$\pm\left(\frac{\sigma_1}{\sigma_{fs}}-\frac{\sigma_2}{\sigma_{fs}}\right)=\frac{4}{9}\left(\frac{t}{l}\right)^2\left\{1-\frac{3\sqrt{3}}{4\left(\frac{t}{l}\right)}\left(\frac{\sigma_1}{\sigma_{fs}}+\frac{1}{3}\frac{\sigma_2}{\sigma_{fs}}\right)\right\} \quad (1)$$

The values of the biaxial stresses were taken to be three times the supply pressure [44]. The other constraint was supplied by Eqn. (2), which ensures contact between hexagonal HTE network and the reactor inner surface whilst leaving room for the porous filter at the centre as shown in Fig. 1. This yields a honeycomb HTE network of thickness (t) 0.9795 cm and length (l) 2.7438 cm which is in consonance with the safe dimension values suggested by Visaria [45]. Sectioned view of the reactor assembly is depicted in Fig. 2.

$$R_i^2 = 4.3329 l_2^2 + l_1^2 + 4 l_1 l_2 \quad (2)$$

where R_i is the inner radius of the cylindrical reactor, l_1 and l_2 are the distances of the inner and outer edge of the hexagon from the centre, respectively. The HTE location was optimized by conducting a parametric study separately, placing the HTE network at distances of 1 cm, 2 cm, 3 cm and 4 cm. On the basis of the observations made, the optimum location was chosen.

Mathematical modelling

The absorption and desorption processes taking place in the reactor were simulated using COMSOL 5.2. A 3-D

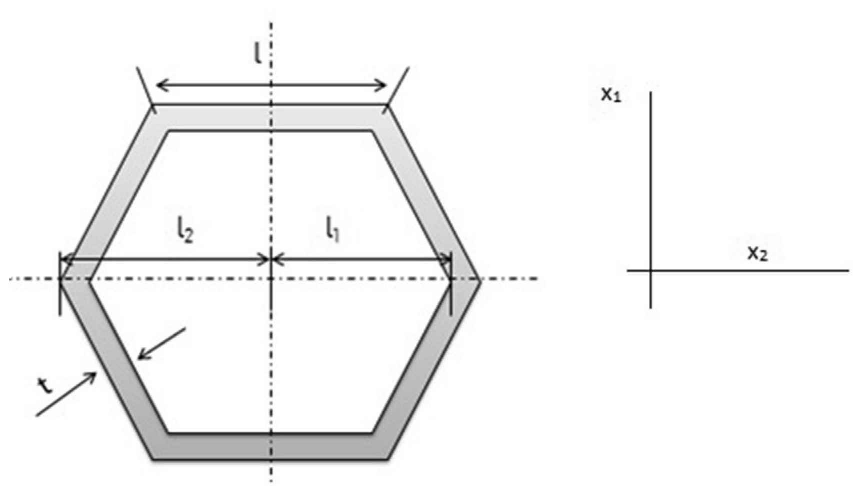

Fig. 1 – Hexagonal honeycomb cell.

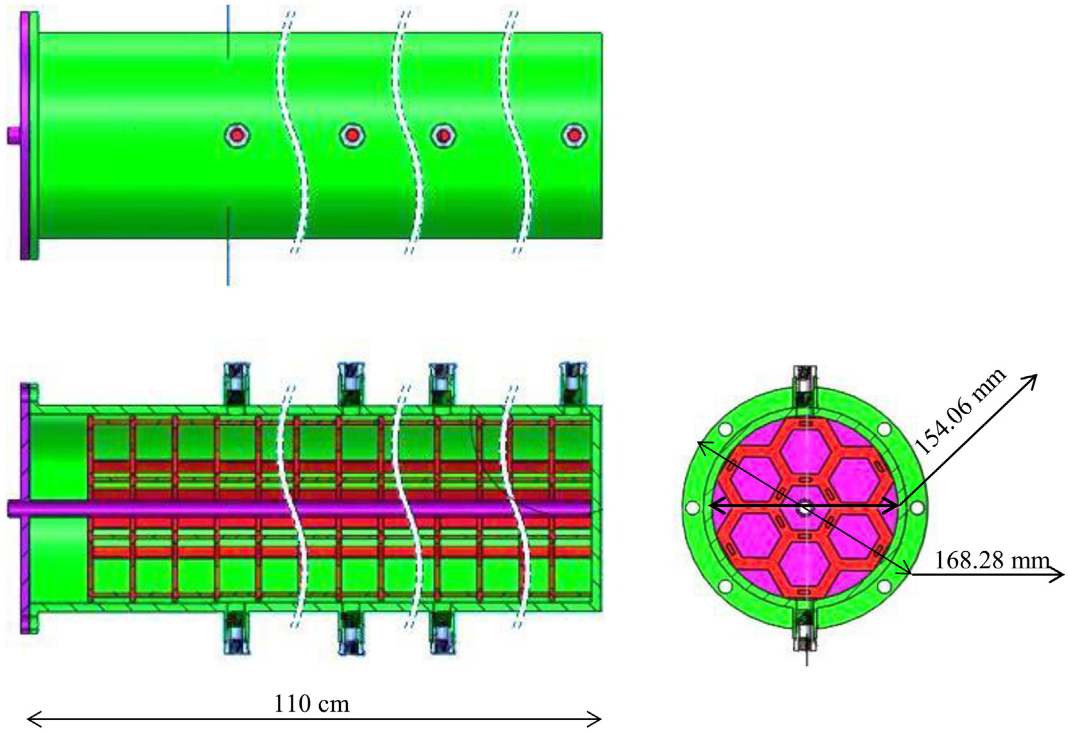

Fig. 2 – Diagram showing the assembled and sectioned view of the reactor.

mathematical model was developed to study the processes under varying operating conditions. The model is based on previous studies conducted on cylindrical reactors [46–48].

The following assumptions have been used for modelling the system:

1) The solid and gas are at the same temperature (local thermal equilibrium).
2) Gas phase: hydrogen behaves as an ideal gas.
3) Solid phase: MH is isotropic and has uniform porosity.
4) The pressure inside the tank is uniform throughout.

The simulation domain consists of a slice of the metal hydride bed, SS316L tank, and the aluminium HTE network. The reactor is exposed to the heat transfer fluid at different conditions. The alloy considered for the model is AB_5 type, $La_{0.9}Ce_{0.1}Ni_5$, chosen for its good kinetics and reversibility, and fuel cell friendly operating temperature and pressure. The details of governing equations and boundary conditions are as follows:

Continuity

The continuity equation for the expansion volume part which is the gas filled void space, and the metal hydride bed is represented by Eqn. (3). Here ρ_g is the density of hydrogen gas and ρ_m is the density of the metal hydride. The density of the metal hydride depends upon M which is the time rate of absorption/desorption of hydrogen per unit volume. Note that the M will be negative in Eqn. (3) and positive in equation (4) for absorption and vice versa for desorption.

$$\text{MH bed: } \varepsilon\frac{\partial\rho_g}{\partial t}+\nabla.(\rho_g.\mathbf{u}) = -M \tag{3}$$

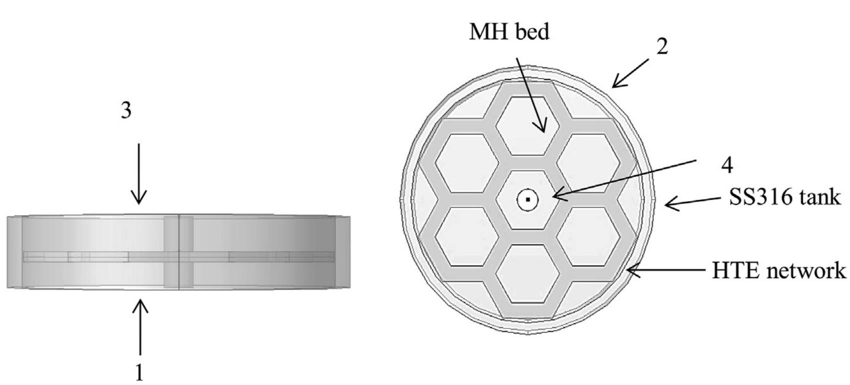

Fig. 3 – Boundary definition for the simulation domain.

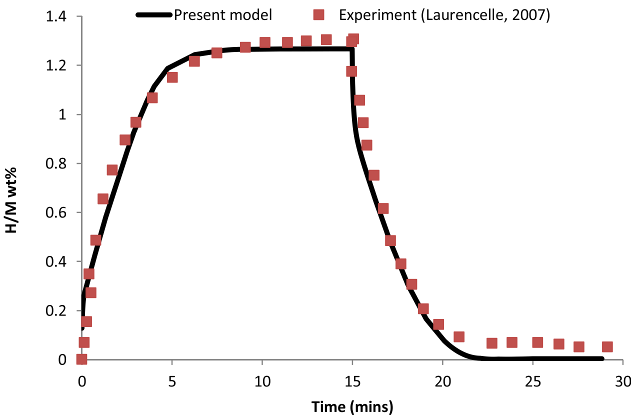

Fig. 4 – Comparison of H/M weight per cent from the present model and experimental data.

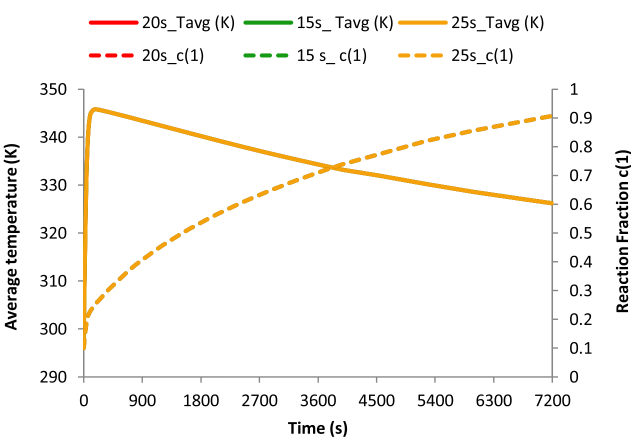

Fig. 5 – Time step independence test for mathematical model for MHHSR equipped with hexagonal honeycomb HTE.

$$(1-\varepsilon)\frac{\partial \rho_m}{\partial t} = M \quad (4)$$

Conservation of momentum

The conservation of momentum is described by the Darcy's law. Here μ_g, ε, and κ are the viscosity of the gas, the porosity and the permeability of the bed respectively; F is the body force. The variable $\boldsymbol{u}$ represents the velocity of hydrogen gas within the MH bed.

$$\text{MH bed: } u = -\frac{\kappa}{\mu_g}\nabla \boldsymbol{p} \quad (5)$$

Conservation of energy

Assuming thermal equilibrium between gas and solid, T represents temperature within the system. The variables C_{pg}, k_g and k_m are specific heat of the gas, thermal conductivity of the gas and thermal conductivity of the porous bed respectively. The effective thermal conductivity and the effective thermal mass of the porous bed and the gas is represented by k_e and $(\rho C_p)_e$.

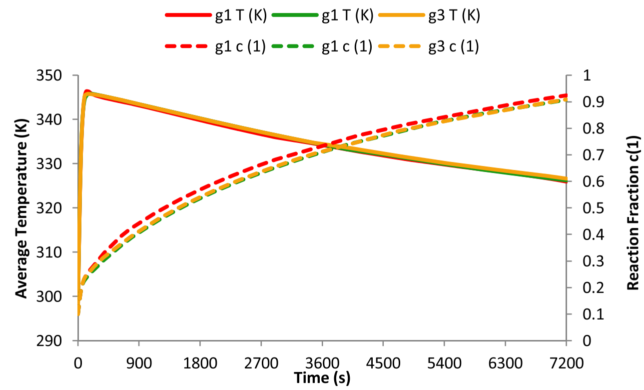

Fig. 6 – Grid independence for mathematical model of MHHSR equipped with hexagonal honeycomb HTE.

Table 1 – Grid properties.

S.no.	Grid	No. of elements	Max. element size (mm)	Min. element size (mm)
1.	g1	175,836	4.9	1.37
2.	g2	284,217	4.76	1.42
3.	g3	310,111	3.5	1.25

Table 2 – Thermophysical properties of the metal alloy.

Parameter	Value
Alloy density	8527 kg/m^3
Alloy specific heat	360 kJ/kg
Alloy thermal conductivity	1 W/mK
Porosity	0.5
Activation energy (absorption)	28,540
Activation energy (desorption)	28,260 J/mol
Enthalpy of reaction (absorption)	26,562 J/mol
Enthalpy of reaction (desorption)	27,960 J/mol
Entropy (absorption)	104 J/molK
Entropy (desorption)	106 J/mol K
Reaction constant (absorption)	59 s^{-1}
Reaction constant (desorption)	9.57 s^{-1}

$$\text{MH bed: } (\rho C_p)_e\frac{\partial T}{\partial t} + \rho_g C_{pg}(u.\nabla T) = \nabla.(k_e \nabla T) + Q_s \quad (6)$$

where,

$$(\rho C_p)_e = \varepsilon \rho_g C_{pg} + (1-\varepsilon)\rho_m C_{pm} \quad (7a)$$

$$k_e = \varepsilon k_g + (1-\varepsilon) k_m \quad (7b)$$

$$\text{HTE network: } (\rho C_p)_f\frac{\partial T}{\partial t} + \rho_f C_{pf}(u.\nabla T) = \nabla.(k_f \nabla T) \quad (8a)$$

$$\text{Steel tank: } (\rho C_p)_{steel}\frac{\partial T}{\partial t} + \rho_{steel} C_{psteel}(u.\nabla T) = \nabla.(k_{steel} \nabla T) \quad (8b)$$

The reaction kinetics that determine the amount of hydrogen absorbed/desorbed are expressed as:

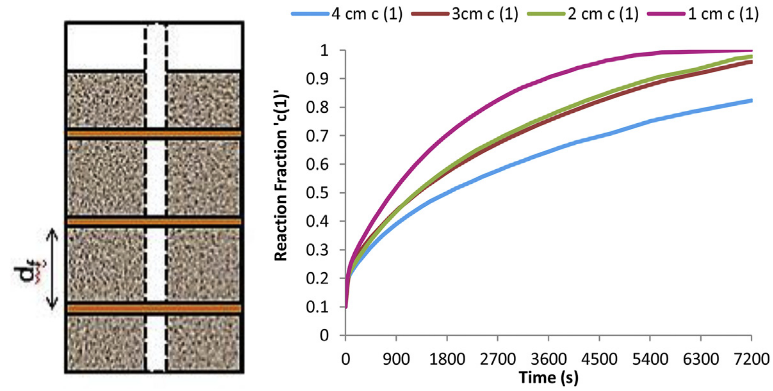

Fig. 7 – Influence of HTE distance on hydrogen absorption.

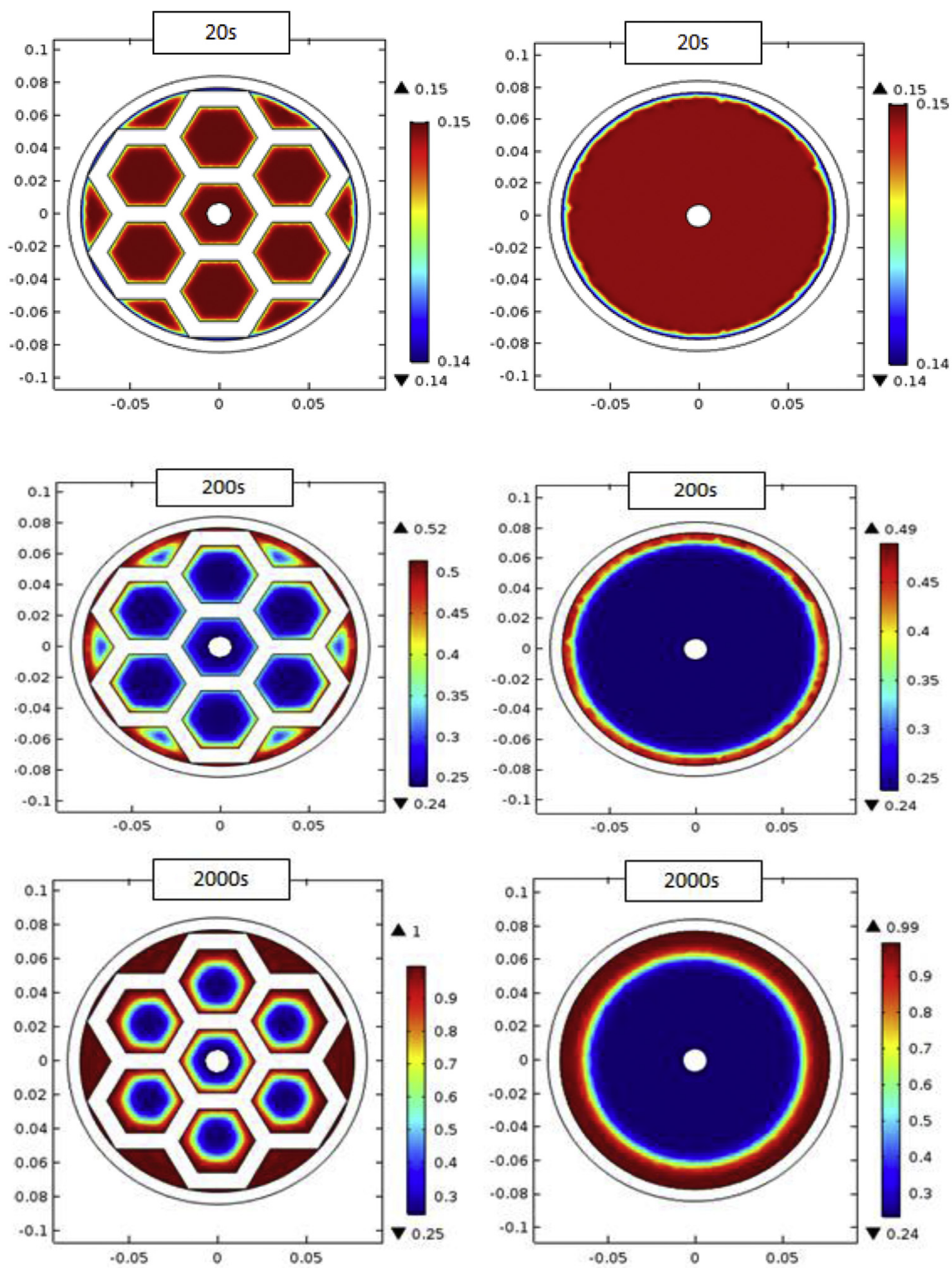

Fig. 8 – Comparison of reaction fraction evolution of reactor with hexagonal HTE network and reactor without any HTEs.

$$\text{Absorption: } M = C_a \exp\left(\frac{-E_a}{R_g{*}T}\right)\ln\left(\frac{p}{P_{eqa}}\right)(\rho_{ss} - \rho_m) \tag{9a}$$

$$\text{Desorption: } M = C_d \exp\left(\frac{-E_d}{R_g{*}T}\right)\left(\frac{p - P_{eqd}}{P_{eqd}}\right)(\rho_m - \rho_e) \tag{9b}$$

where,

$$\frac{P_{eq}}{P_{ref}} = \left(exp\left(\frac{S}{R} - \left(\frac{H}{R{*}T}\right)\right) + 0.51 * tan\left(pi * \left(c - 0.5\right)\right) + 0.15\right) \tag{10}$$

The equilibrium pressures are predicted with the help of van't Hoff equation as stated in Eqn. (10). Please note that 'c' is the fraction of hydrogen absorbed/desorbed.

Initial and boundary conditions

For absorption the initial temperature of the tank is considered to be 298 K and the pressure is 25 bar which is well above the equilibrium pressure of the alloy under consideration. Along the tank wall, a heat flux condition is set with constant heat transfer coefficient. A no-slip boundary condition is set along the tank walls except for the inlet through which hydrogen is allowed to enter. Continuity of temperature, density, velocity and other properties is established at the boundaries between the MH bed and the reactor body and the HTE network. The boundary and initial conditions may be summarized as follows:

$T\ (r,\theta,z,0) = 298$ K; $p\ (r,\theta,z,\ 0) = 25$ bar; $\boldsymbol{u}\ (r,\theta,z,0) = 0$;
$\rho_m\ (r,\theta,z,\ 0) = 8400\ \text{kg/m}^3$;

$$\rho_g(r,\theta,z,0) = \frac{p_g{*}R_s}{T}$$

As can be seen from Fig. 3, boundaries 1 and 3 are assumed to be thermally insulated, while 2 is subjected to a heat flux through the cooling/heating fluid. Boundary 4 is exposed to the inlet pressure.

$$\text{At 1 and 3: } -k\frac{\partial T}{\partial n} = 0;\ \boldsymbol{u} = 0$$

$$\text{At 2: } -k\frac{\partial T}{\partial n} = h(T - T_f)$$

$$\text{At 4: } n.\nabla T = 0; \quad p = p_{inlet/outlet}$$

Model validation

The model is based on previous studies conducted on cylindrical reactors [46–48]. It has been validated in the present study as shown in Fig. 4. The results are matched with the study by Laurencelle and Goyette [49] and are found to be in good agreement.

Grid and time step independence

From Fig. 5 it can be clearly observed that the proposed mathematical model is independent of the time step taken for solving the model. The model was tested for three different time steps viz 15s, 20s, and 25s. The performance curves depicting fraction of hydrogen absorbed and average bed temperature from all the three cases are completely concurrent.

Fig. 6 depicts the grid independence characteristics of the model. The model was tested using three different tetrahedral grids: g1 containing 175,836 tetrahedral elements, g2 containing elements 284,217 and g3 containing 310,111

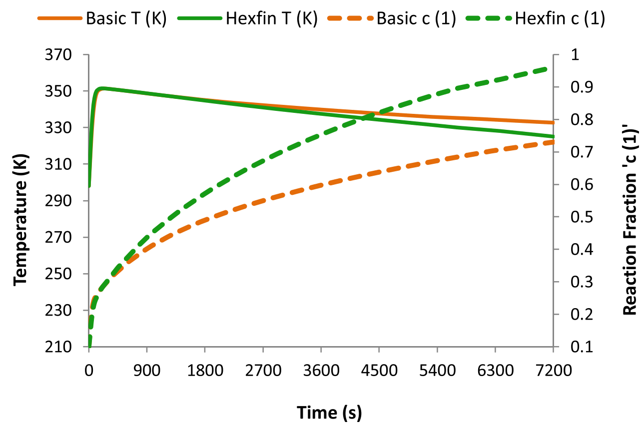

Fig. 9 – Comparison of average reaction fraction and average bed temperature in the reactor with hexagonal honeycomb HTE (hexfin) and a reactor with no HTE (basic).

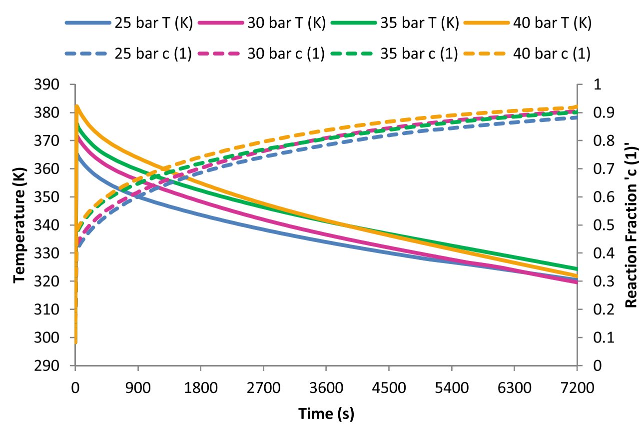

Fig. 10 – Effect of pressure on average bed temperature and average reaction fraction during absorption.

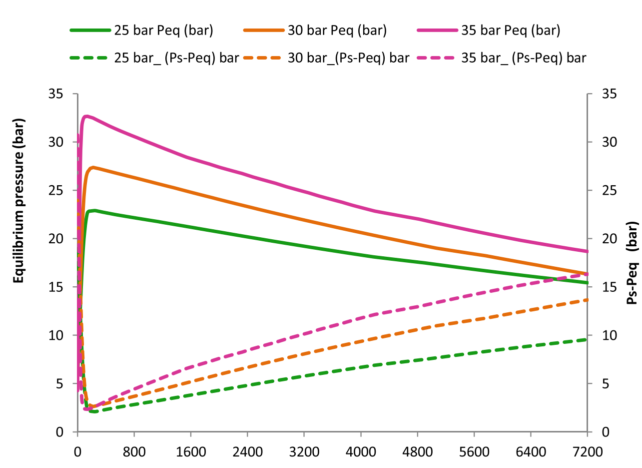

Fig. 11 – Effect of increase in pressure on equilibrium pressure evolution.

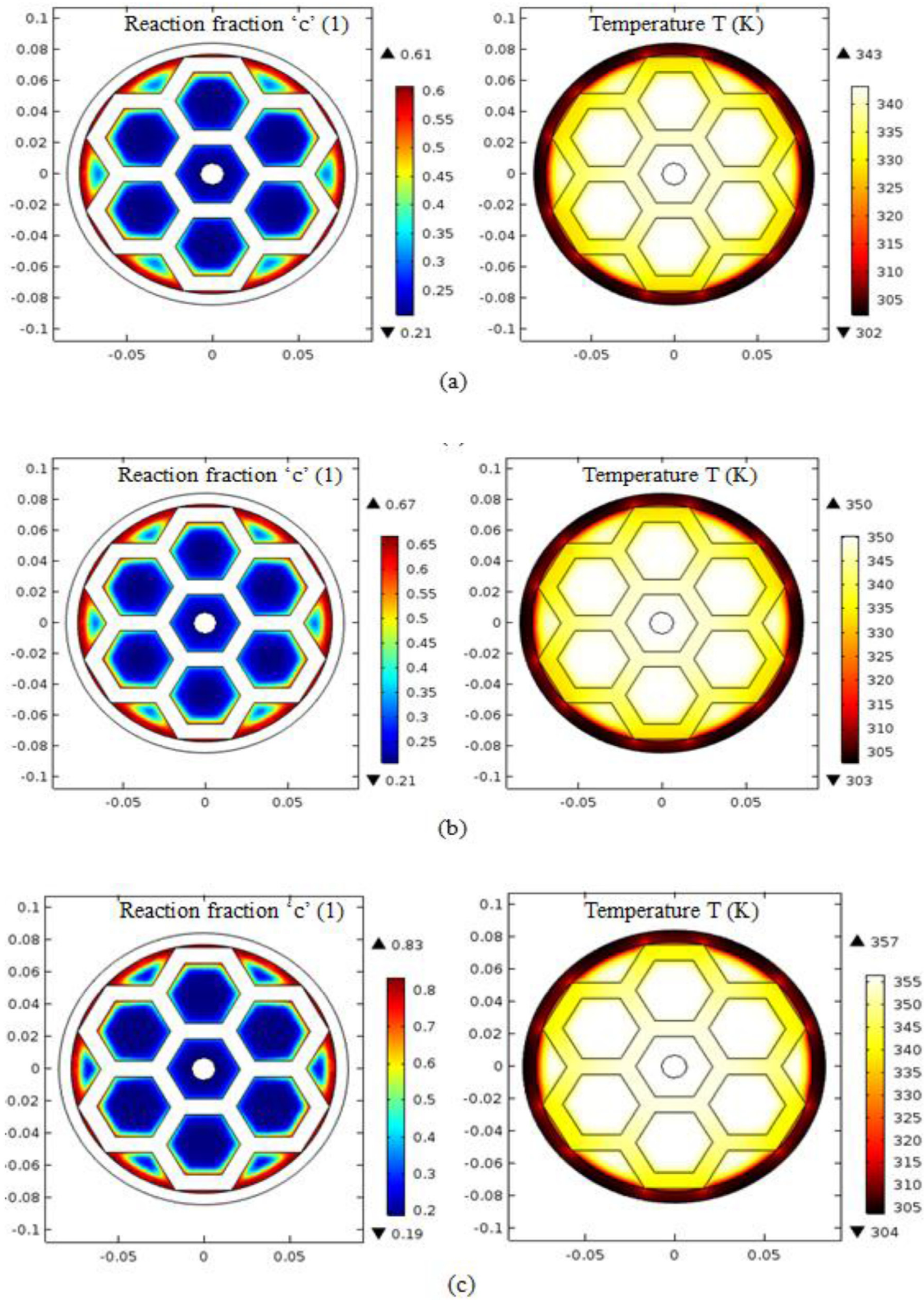

Fig. 12 – Variation of reaction fraction and bed temperature at 500 s at a) 25 bar b) 30 bar c) 35 bar.

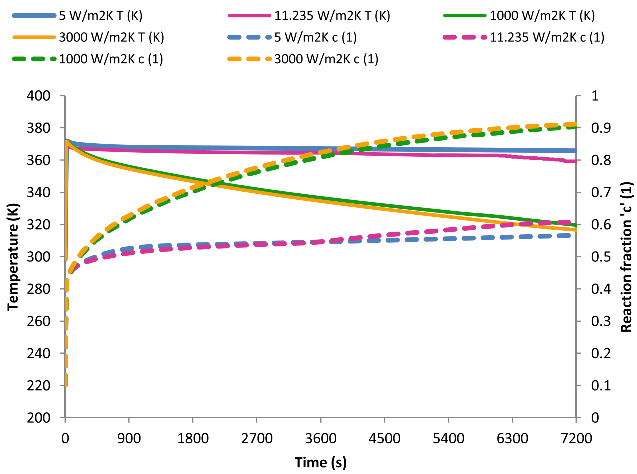

Fig. 13 – Effect of heat transfer coefficient on average bed temperature and average reaction fraction during absorption.

tetrahedral elements. The performance curves depicting fraction of hydrogen absorbed and average bed temperature do not show any significant difference. The numerical analysis was performed using grid g2. Table 1 summarizes the properties of the grids used.

Results and discussions

The reactor was modeled using COMSOL 5.2 and the absorption and desorption processes were analysed under different operating conditions such as supply pressure, convection conditions, heat transfer fluid temperature. The thermo-physical properties of $La_{0.9}Ce_{0.1}Ni_5$ are given in Table 2 [50,51].

Optimization of HTE location

The HTE location is optimized after modelling the performance of the reactor by placing the HTEs at different locations along the reactor axis. The locations were at distances of 1 cm, 2 cm, 3 cm, 4 cm along the reactor axis. Hydrogen at a supply pressure of 25 bar was injected into the reactor. The evolution of reaction fraction completed i.e. fraction of hydrogen absorbed with time was studied.

It was noted that the reactor performance was distinctly superior when the HTEs were placed at a distance of 3 cm from each other. The rate of hydrogen absorption was significantly higher than when the HTEs were 4 cm apart. When the distance was further decreased, the improvement in rate of reaction was not very significant as shown in Fig. 7. From these observations, axial distance between two consecutive HTEs was fixed at 3 cm.

Absorption

The absorption reaction occurs when hydrogen at high pressure is injected into the reactor. If the supply pressure is greater than the equilibrium pressure at that temperature, hydrogen will get absorbed into the alloy forming a hydride. Since this reaction is highly exothermic, heat removal from the MH bed plays a key role in ensuring that the reaction continues to progress.

Fig. 8 compares the reaction fraction and bed temperatures across the MHHSR cross-section with the hexagonal HTE network to a reactor with no heat transfer enhancements at a. pressure of 35 bar. A major difference can be noted in the hydrogen absorption rates between. the two with the HTE equipped reactor exhibiting significantly higher hydrogen absorption than the basic reactor with no enhancements. Since this reaction is highly exothermic, heat removal from the MH bed plays a key role in ensuring that the reaction continues to progress. From Fig. 8 it is evident that not only is the maximum reaction fraction completed, higher in the reactor with hexagonal HTE network, it is spread over a wider area indicating the suitability of the HTE network as an effective conduction pathway. After 200 s, the hexagonal HTE reactor has completed a maximum of 52% of the reaction whereas in the reactor with no HTE network, the maximum reaction fraction is completed is less than 50% besides being spread over a much smaller area.

Fig. 9 compares the average reaction fraction and average bed temperature in a reactor with no HTEs (basic) and the present MHHSR (hexfin). The supply pressure is 35 bar whereas the overall heat transfer coefficient is 1112.898 W/m^2K and the ambient temperature is 298 K. It can be observed that the proposed MHHSR equipped with hexagonal honeycomb HTE (hexfin) absorbs hydrogen at a significantly faster rate as compared to the MHHSR with no HTE (basic). The reactor equipped with hexagonal HTEs attains 90% reaction rate at 5820 s, an improvement of more than 30% over the basic reactor. In the initial phase of the reaction, reaction fraction as well as the bed temperature in both the cases is equivalent, however, as the reaction progresses, the proposed MHHSR exhibits faster temperature drop and hydrogen absorption. This is due to the fact that in the initial phase of the reaction, the hydrogen absorption rate has a weak dependence on heat transfer from the alloy bed; reaction kinetics plays a more important role in determining the reaction rate. As the reaction progresses, heat transfer becomes more important and hence, the proposed MHHSR shows an improved rate of heat transfer and hydrogen absorption [52,53].

Influence of operating parameters

Effect of supply pressure. Since the driving force of the reaction is the difference between the supply pressure and the equilibrium pressure; with increase in supply pressure, the rate of reaction is observed to improve. In the initial phase of the reaction, the absorption occurs rapidly because the difference between the supply pressure and the equilibrium pressure is the greatest, considering the reactor to be at room temperature. As the exothermic absorption reaction progresses the bed temperature rises as can be observed in Fig. 10. This causes the equilibrium pressure to rise correspondingly, leading to a decrease in reaction rate. Even though over 90%

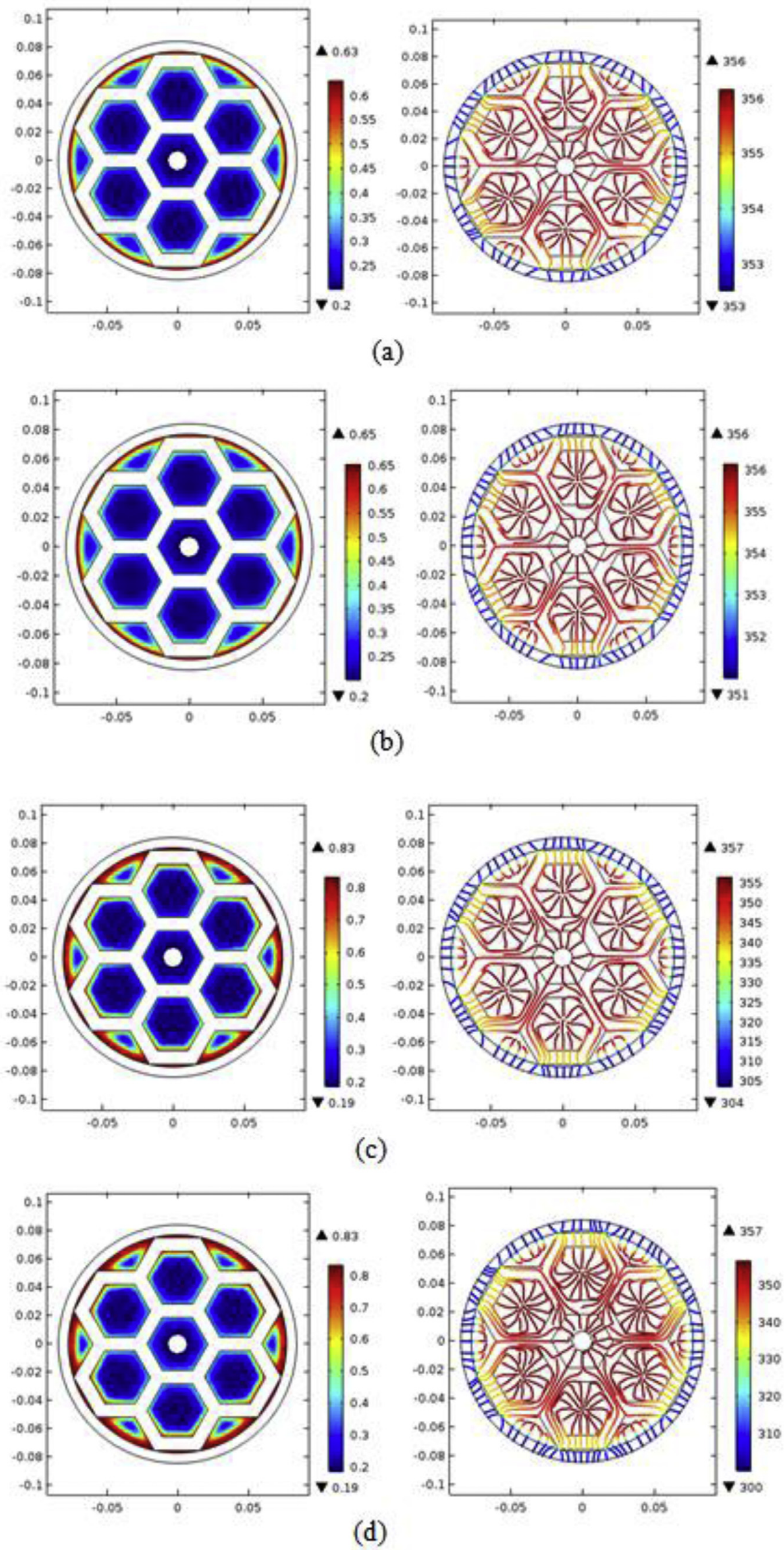

Fig. 14 – Variation of reaction fraction and heat flux streamlines across the MHHSR cross-section at 500 s with varying heat transfer coefficient a) 5 W/m²K b) 10 W/m²K c) 1112.898 W/m²K d) 3000 W/m²K.

absorption is achieved in all the cases. The 90% reaction fraction is achieved at 5600 s at 40 bar, whereas under 25 bar pressure it is reached at around 7200 s, increasing the reaction rate by around 22%. The rate of reaction is therefore, higher for higher supply pressure. The average bed temperature is also seen to increase with increase in supply pressure. The temperature and rate of reaction have a cyclic relationship. As the rate of reaction increases, the bed temperature also increases more rapidly; however, this increase in temperature in turn causes the reaction to slow down unless the heat is withdrawn from the MH bed. In the initial phase, the rate of reaction and hence the rate of heat generation is much higher than the rate of heat removal from the MH bed, hence the rapid shoot in bed temperature.

Fig. 11 shows the variation in equilibrium pressure with supply pressure. With increase in supply pressure, there is an overall increase in equilibrium pressure. The equilibrium pressure shoots rapidly as the reaction progresses due to increase in bed temperature and then drops slowly. The difference in equilibrium pressure and supply pressure which drives the reaction, is also observed to increase with increasing supply pressure thus explaining the faster absorption at higher pressures.

Even though the supply pressure is the same throughout the volume of the reactor, the equilibrium pressure varies spatially, because the temperature is different at different locations in the reactor depending upon the reaction rate and heat transfer. This in turn leads to further variation in hydrogen absorption rate as they share a cyclic relationship. Fig. 12 depicts the hydrogen absorption completed (reaction fraction) and bed temperature at different pressures across the MHHSR cross-section. The maximum fraction of hydrogen absorbed across the alloy bed increases with increase in supply pressure. At 500 s the maximum reaction fraction at 25 bar is 0.61; at 35 bar there is an increment of 36% in the same. The variation of bed temperature and reaction fraction can also be compared at each pressure from Fig. 12. In the areas closer to wall and the aluminium HTE network, the temperature is lower and consequently, the reaction fraction is higher (because of comparatively low equilibrium pressure).

Effect of heat transfer coefficient. Fig. 13 shows the variation of hydrogen absorption rate and average bed temperature of the reactor under different convection conditions. The supply pressure is taken as 35 bar and the heat transfer fluid temperature is constant at 298 K, while the heat transfer coefficient is varied. The heat transfer coefficients which correspond to natural convection in air, forced convection in air, natural convection in water and forced convection in water are 5 W/m^{2}K, 11.235 W/m^{2}K, 1112.898 W/m^{2}K and 3000 W/m^{2}K. It is observed that there is a significant improvement in the hydrogen absorption rate when the process takes place at a high heat transfer coefficients of 1000 W/m^{2}K and 3000 W/m^{2}K i.e. with water as heat transfer fluid. When the heat transfer coefficient is increased from 5 to 11.235 W/m^{2}K i.e. when the process is carried out with forced convection in air instead of natural convection no significant improvement is registered. Nor does increasing the heat transfer coefficient from 1112.898 W/m^{2}K to 3000 W/m^{2}K lead to any noteworthy improvement in hydrogen absorption.

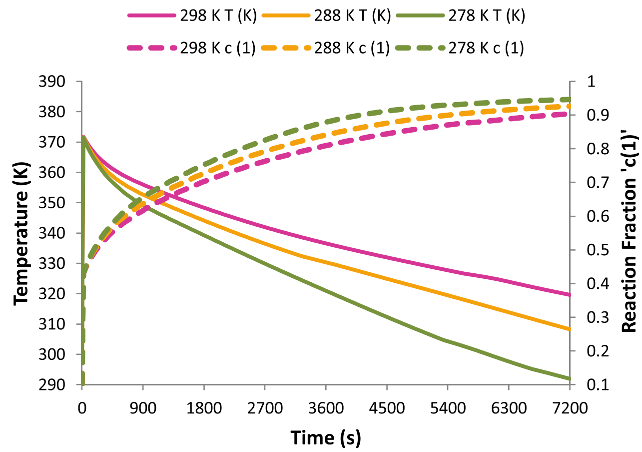

Fig. 15 – Effect of HTF temperature on average bed temperature and average reaction fraction during absorption.

Across the reactor cross-section, the effect of varying the heat transfer coefficient can be observed from Fig. 14. The reactor slices depicting reaction fraction and heat flux streamlines are recorded at 500 s. The maximum reaction fraction attained at a heat transfer coefficient of 5 W/m^2 K is 0.63 confined to the area very close to the wall. The case of forced convection in air with heat transfer coefficient 10 W/m^{2}K registers a slight increase in the maximum reaction fraction which is 0.65. Once the heat transfer coefficient is increased to 1112.898 W/m^{2}K, the reaction fraction registers a 27% increase and maximum reaction fraction is 0.83. However, further increment in heat transfer coefficient to 3000 W/m^{2}K does not lead to any improvement in maximum reaction fraction as was observed in Fig. 14 for average reaction fraction. From the heat flux streamlines it can be observed that for each case the heat flux streamlines originate at the centre of the hexagonal cell and travel to the MHHSR periphery through the hexagonal HTE where the heat transfer fluid takes up the heat.

Effect of heat transfer fluid temperature. The effect of heat transfer fluid temperature on the rate of hydrogen absorption is shown in Fig. 15. At 35 bar supply pressure, the heat transfer fluid temperature is varied from 298 K to 278 K in

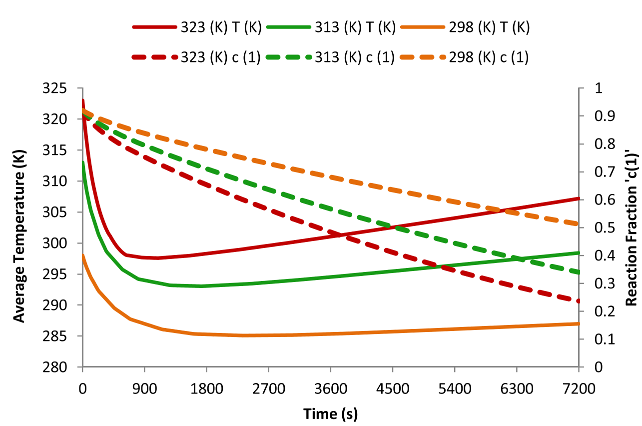

Fig. 16 – Variation of average bed temperature and average reaction fraction for desorption.

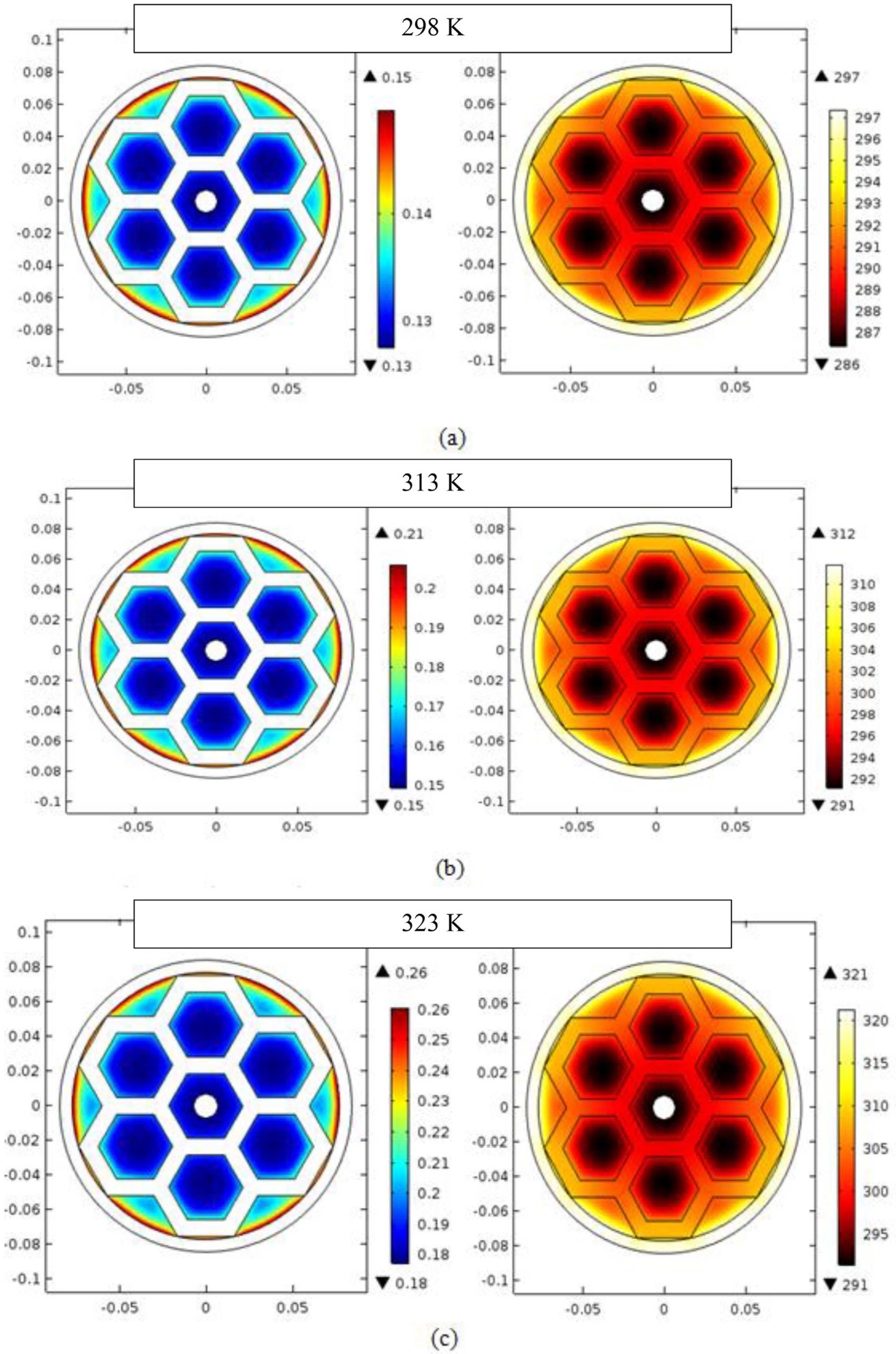

Fig. 17 – Reaction fraction and average bed temperature with variation in heat transfer fluid temperature.

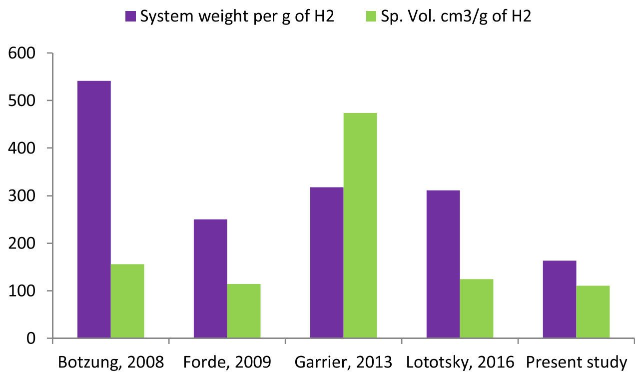

Fig. 18 – Comparison of gravimetric and volumetric efficiencies of a few large scale reactors.

steps of 10 K. Decrease in heat transfer fluid temperature increases the rate of heat removal from the MH bed via convection, and increases the rate of hydrogen absorption; although the increase in hydrogen absorption is not very remarkable. In case of the HTF temperature of 278 K, the decrease in average bed temperature occurs at a much faster rate as compared to when the temperature is 288 K and 298 K but there is only a slight effect on the reaction rate. These results are in agreement with the sensitivity tests performed on large scale reactor [53].

Desorption

The desorption reaction in the MHHSR was modeled by assuming outlet pressure as atmospheric pressure. Since desorption reaction is an endothermic reaction, a constant supply of heat has to be supplied in order to maintain the reaction equilibrium. In the present study, a constant temperature heat source is provided along the external surface area of the reactor. Mathematically, a convective heat transfer boundary condition assuming water under natural convection was taken. The performance of the system was analysed under different heat transfer fluid temperatures: 298 K i.e. room temperature, 313 K, and 323 K and is shown in Fig. 16. As the external temperature is increased, the bed temperature rises and so does the equilibrium pressure. Since the equilibrium pressure is a direct function of the temperature, therefore, increase in temperature results in increase in equilibrium pressure. This increases the driving potential, i.e. the difference between desorption pressure (equilibrium pressure) and the discharge pressure (ambient pressure), and hence the rate of hydrogen desorbed.

Since desorption is an endothermic reaction, the heat of the reaction is taken up from the surroundings. When the surrounding temperature is high, the reaction is facilitated and the rate of desorption can be observed to be higher as depicted in Fig. 17. The maximum and minimum reaction fraction desorbed is seen to increase with increased heat transfer fluid temperature. The temperature profile across the cross-section shows minimum value of temperature at the core of the reactor and higher temperature at the circumference. The reaction at the core progresses relatively slowly corresponding to the low temperature, whereas the areas away from the core and closer to the external heat transfer fluid exhibit faster reaction rate.

Comparative analysis

While metal hydride based hydrogen storage systems are known for their high volumetric efficiencies, low system gravimetric capacities tend to pose a major challenge especially in case of mobile applications. Fig. 18 compares the system weight per unit weight of H_2 for different large scale systems developed in the last few years [10,29,54,55] with the design proposed in the present study. Botzung et al. employed a plate-fin type reactor [10] whereas Førde et al. used a u-shaped copper tube with fins to enhance the heat transfer characteristics of the MHHSR [54]. Garrier et al. employed a higher capacity Mg based alloy for hydrogen storage accompanied with $Mg_{69}Zn_{28}Al_3$ as Phase Change Material (PCM) which acts as a heat sink/source. Lototsky et al. used MHHSR equipped with copper fins as heat transfer enhancements. Although, it is difficult to conclusively compare systems on the basis of MHHSR design alone, since performance is a strong function of operating parameters such as supply pressure, heat transfer fluid temperature etc. However, the present design enables lowering of system weight to a great extent without affecting the system performance significantly. The HTE proposed in the present design has an inherently high surface area to volume ratio, enabling high heat transfer to/from the MHHSR without adding redundant system weight. From Fig. 18, it can be observed that the hexagonal honeycomb HTE based single reactor design allows for a significant reduction in system weight while offering a comparable volumetric efficiency.

Conclusions

In this article a novel heat transfer mechanism based on hexagonal honeycomb heat transfer enhancements (HTE) is proposed for a metal hydride hydrogen storage reactor. The impact of the HTE network on the performance of the metal hydride storage reactor during absorption and desorption is studied numerically under different operating conditions of pressure, heat transfer coefficient and heat transfer fluid temperature. It is observed that the addition of hexagonal HTE network improves the absorption performance of the hydride bed by over 30%. A 22% increment in absorption rate is noted when the supply pressure is increased from 25 bar to 40 bar. Increasing heat transfer coefficient from 5 to 11 W/m^2K does not result in any significant improvement in the reaction rate, however when the heat transfer coefficient is increased to 1112.898 W/m^2K there is a remarkable increase in reaction rate. Interestingly, further increase in heat transfer coefficient does not impact the rate of absorption. Decrease in heat transfer fluid temperature does not have a significant impact on system performance. Desorption studies indicate that higher heat transfer fluid temperatures result in improved desorption rates with 20% improvement from 313 K to 323 K. A comparison of existing large-scale systems with the present system in terms of gravimetric and volumetric efficiencies is presented.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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